Strategic Resources That Matter - a Civilization VII mod

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Strategic Resources That Matter

A Civilization VII mod. Horses, Ivory, Iron, Niter, Hardwood, Oil, Coal and Rubber matter again. A unit that draws on one of them fights weaker and trains slower while its owner has none of it, and each one you do hold still adds the game's own bonus, so the count keeps mattering. Lose your last one, to a raid on the mine or a lost settlement, and the penalty is back a turn later. Past six, where the game's own bonus stops, the count keeps adding strength.



Figure 1: The unit panel of a Chariot with no Horses

A Chariot with no Horses: 27 Combat Strength instead of 30, and the last cell in its stats says why.

Single-player. In multiplayer, losing a resource does not bring the penalty back until that player gains a resource.

What the player sees

Nothing is forbidden. Every unit can still be trained and upgraded as in the base game. What changes is the price of going without:

- **None of the resource:** the unit has less Combat Strength (melee, and ranged and bombard where it has them), and a city puts less Production toward it, so it takes longer to train. With the default setting that is -3 Combat Strength and a third less Production, which makes the unit take half as long again.
- **One or more:** no penalty. The game's own +1 Combat Strength for each applies as before, up to six.
- **More than six:** +1 more Combat Strength at 8, 10, 12, 14, 16, 18 and 20, on the same units the game's own bonus covers.

The game tells you why, in your language:

- **Unit panel:** when you select a unit that is short of its resource, its stats gain one more cell, the resource's icon over the penalty in red; its tooltip says ”Unsupplied: no Iron (-3 Combat Strength, slower to train)”.
- **Combat preview:** the penalty is a line of its own beside the game's modifiers, for your unit and for the enemy's: ”-3 Unsupplied: no Niter”. Past six, the extra strength shows as one line beside the game's own, such as ”+1 Niter: 8 or more”.
- **Production list:** a unit that needs a resource carries a pill on its item, beside the game's own text: ”Needs Niter” on a red tint while you have none (the turn count shown is the slower one), ”Niter: 2” once you have some. The item's hover tooltip carries the whole sentence under the unit's description: ”Needs Niter. Without it: -3 Combat Strength, slower to train.” The pill follows a resource gained or lost while the list is open.
- **Resources:** a strategic resource's description, on the map and in the commerce screen, ends with the units it supplies for you in this age: ”Supplies units: AT Gun, Assault Gun, Field Cannon and 3 more.” A resource that doesn't supply any unit in this age keeps the game's own description.
- **Civilopedia:** a page under Game Concepts, Combat, ”Strategic Resources”, with the rules and the table below; and on each gated unit's page a line such as ”Needs Iron: with none, -3 Combat Strength and slower to train”, with the penalty your game was set up with.

- **Notification:** when you lose the last of a resource your units need, one notification at the start of the next turn: "Supply lost: no Niter. Your Field Cannon and 3 other units fight at -3 and train slower until you get Niter back." Once per loss; it ends with the turn, never holds the turn up, and a right-click dismisses it. Two losses in one turn share one notification.
- **The first time:** the first time you select one of your units while it is short of its resource, an advisor callout explains the rule in two sentences, and "Tell me more" opens the Civilopedia page. It shows once, not in every game, and only with tutorials or advisor warnings on in the game's options.

Which units need what

A civilization's unique unit needs what the unit it replaces needs.

Age	Units	Resource
Antiquity	Cavalry: Chariot, Horseman and their unique versions	Horses
Antiquity	War elephants: the Khmer's Yuthahathi and the Maurya's Purabhettarah	Ivory
Antiquity	Infantry from the second tier: Spearman, Phalanx, Legion, Immortal and the others	Iron
Exploration	Cavalry: Courser, Knight, Lancer and their unique versions	Horses
Exploration	Infantry from the second tier: Man-At-Arms, Pikeman and their unique versions	Iron
Exploration	Siege from the third tier: Bombard and its unique version	Niter
Exploration	Heavy warships from the second tier: Carrack, Galleon and their unique versions	Hardwood
Modern	Field artillery: Field Cannon, Field Gun and their unique versions	Niter
Modern	The heavier artillery: Mortar, Howitzer, AT Gun, Assault Gun and their unique versions	Oil
Modern	Armour: Landship, Tank and their unique versions	Oil
Modern	Battleships: Dreadnought, Battleship and their unique versions	Oil
Modern	Light warships: Ironclad, Cruiser, Destroyer, Barbary Corsair	Coal
Modern	The Cuirassier and the unique units that replace it (Cossack, Ghilman, Hussar at their first tier)	Horses
Modern	The Ship of the Line and the Turtle Ship	Hardwood
Modern	The Submarine	Oil
Modern	Aircraft	Rubber

A resource that reaches you over a trade route counts the same as one you mine yourself, from the turn the route lands.

The war elephants count Ivory the way cavalry counts Horses: +1 Combat Strength per Ivory you hold, up to six, and the steps past six with Strategic Resources Past Six on. The game's own +1 per Horses stops at them, so an elephant scales with Ivory alone. Ivory is far commoner than Horses on the map, so the Khmer and Maurya are the best-supplied cavalry in Antiquity; the balance numbers are in docs/elephants-on-ivory.md. With Modern Supply Chains on (a setup option, off by default), every Modern unit needs a second resource as well: see below.

Free of any requirement: the Warrior line and the Scout, ranged infantry (Archer, Crossbowman, Arquebusier), the Catapult and the Trebuchet, the Cog, and Modern infantry, so every player can always defend.

Losing a resource

The game notices a new resource the moment it arrives: a mine finishing on Iron lifts the penalty on your Spearmen at once. It does not notice one going away. The mod makes it check every player again at the start of each turn, so a resource lost by pillage, by a tile changing hands or by anything else brings the penalty back one turn later, for you and for the AI players alike. A trade route ended by a war takes what it brought with it at once; an import that dries up at the other end is gone from your count at once and the penalty follows one turn later. A player who still has one keeps the bonus. The same check keeps the strength past six honest when the count falls, and raises the notification above when your last one goes.

Game setup options

Four options in game setup, beside the game's own options such as age length:

Option	Values	Default
Unsupplied Units	Off; Light (-2, 20% less Production); Standard (-3, 33% less); Harsh (-5, 50% less, twice as long to train); Locked: training; Locked: training and upgrade	Standard
Strategic Resources Past Six	Off; On (+1 at 8, 10, 12, 14, 16, 18 and 20 of a resource)	On
Modern Supply Chains	Off; On (every Modern unit needs a second resource)	Off
Free Tier-Up by Kills	Off; On (the Sengoku's Samurai move up a tier after three kills, Iron or not)	On

Play Now uses the defaults. The options are fixed for the whole game once it starts. Before 1.2 the lock was its own option, Gated Units; a save made with it keeps the gate it was started with.

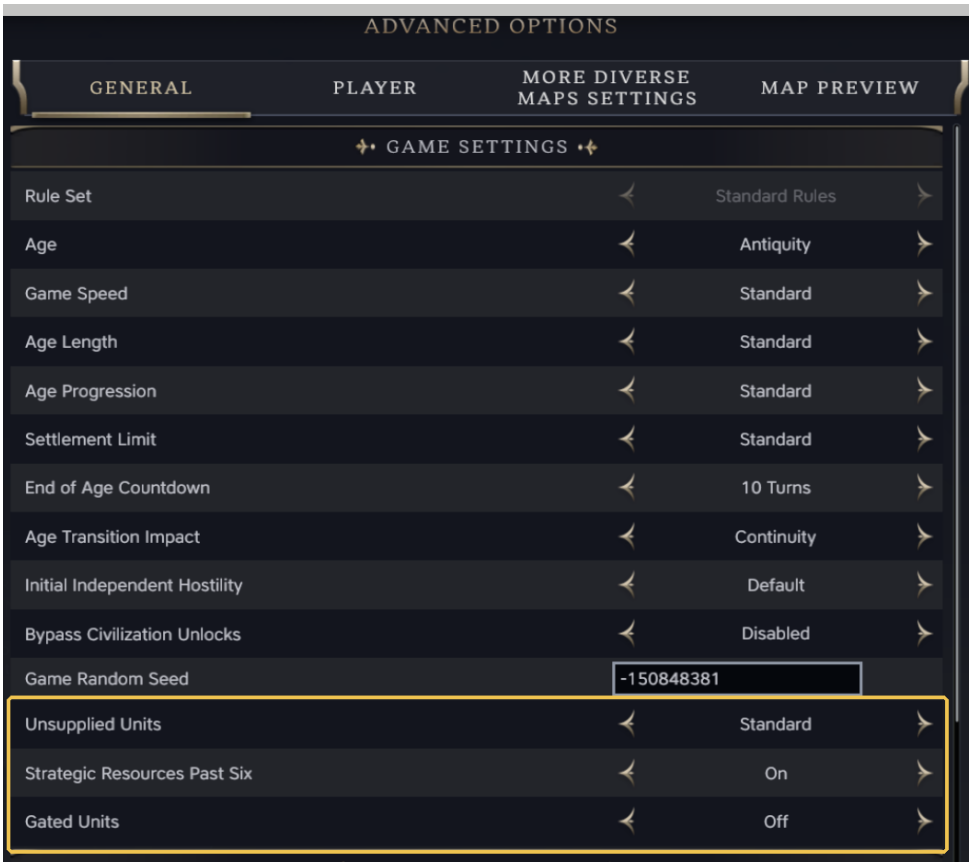


Figure 2: The three options in Advanced Options

Advanced Options, Game Settings: the mod's options sit after the game's own.

Locked units

The two hardest values of Unsupplied Units lock units as well as penalising them. Units in the field take the Harsh penalty, and a unit that needs a strategic resource cannot be trained or bought while you have none of it. It stays in the production list, greyed out, with a ”Locked: no Iron” pill after its stats, until you get some, and it is locked again the moment your last one goes. Under ”Locked: training and upgrade”, no unit can be upgraded into it either, and a commander's Upgrade All leaves it as it is, with the button greyed and the reason in its tooltip, as long as every unit in the army is locked; with a mix, the button stays lit and the tooltip names the units that stay as they are. Under ”Locked: training”, upgrades stay free, as in the base game. Units already in the field are never locked: they keep fighting, at the Harsh penalty, and heal as usual. The units free of any requirement (the Warrior line, the Scout, ranged infantry, the Catapult and Trebuchet, the Cog, Modern infantry) stay free, so every player can always defend. They also stay in the production list when the next unit in their line is gated. In the base game a unit goes out of date as soon as the tech for its upgrade is in, so without Niter you'd lose the Trebuchet at Gunpowder and have no siege unit at all. With Gated Units on, the Trebuchet stays trainable after Gunpowder until you have Niter, and the same goes for the Warrior, the Swordsman, the Cog, the Cuirassier, the Frigate and the unique units in those lines. Once you have the resource, the line works as in the base game: the old unit goes out of date and you can upgrade it. Rome, Persia, Meiji Japan and Great Britain get their own Warrior or Frigate for this, the same unit under the same name and with the same look, which upgrades straight into the Legion, the Immortal, the Mikasa or the Revenge, and takes the same story rewards a Warrior or Frigate would. Independent powers and city-states are never locked. A civilization's unique unit is locked like the unit it replaces.

The Sengoku's Samurai reach their second and third tier by themselves, after three kills at each tier, and the game doesn't ask for Iron for that step, even under ”Locked: training and upgrade”. That's what you get with Free Tier-Up by Kills on, the default. Turn it off under that value and kills don't move a Samurai up any more: it upgrades like any other unit, which needs Iron: tier 2 once you have Heraldry, tier 3 once you have Metal Casting. Off takes out the Samurai's kill counts, so it overrides another mod that changes them. Either way, a Sengoku player with no Iron keeps

the first-tier Samurai after the Kinsei civic, and the second tier takes over once Iron is held. Without the Sengoku DLC the option doesn't change anything.



Figure 3: The production list with a locked Horseman and Legion

Rome with no Horses and no Iron: the Horseman and the Legion stay listed, greyed out, and say why.

The unit panel's cell tooltip ends with "New ones cannot be trained.", the resource description reads "Needed to train: ..." instead of "Supplies units: ...", the lost-supply notification adds "New ones cannot be trained until then.", and a second first-time callout, "Locked unit", explains the lock the first time you select a unit of a locked kind or a city whose list shows one. The Civilopedia page gains a last chapter, "Locked units", and each gated unit's page a line "Cannot be trained or bought without Iron."

Modern Supply Chains

Off by default, because it makes for a hard game. On, every Modern unit that needs a resource needs a second one too, so the supply lines of the industrial age matter as much as the first resource did:

Units	Resources
Field artillery: Field Cannon, Field Gun and their unique versions	Niter and Coal
The heavier artillery: Mortar, Howitzer, AT Gun, Assault Gun and their unique versions	Oil and Coal
Armour: Landship, Tank and their unique versions	Oil and Coal
Battleships: Dreadnought, Battleship and their unique versions	Oil and Coal
Light warships: Ironclad, Cruiser, Destroyer, Barbary Corsair	Coal and Niter
The Cuirassier and its unique versions	Horses and Niter
The Ship of the Line and the Turtle Ship	Hardwood and Niter
The Submarine	Oil and Rubber

Units	Resources
Aircraft	Rubber and Oil

Iron isn't placed on Modern maps, so Coal stands for steel. Each missing resource costs the full Unsupplied Units penalty, so a Tank with neither Oil nor Coal fights at twice it, and the unit panel shows one cell per missing resource; the production pill reads "Needs Oil, Needs Coal". Under a locked value the unit needs both to be trained, bought or (under "training and upgrade") upgraded into, and the pill reads "Locked: no Coal" for whichever is missing. The lost-supply notification, the resource descriptions and the Civilopedia lines follow the second resource too.

AI players

The AI plays under the same rules. The penalty is ordinary unit and city data, so it reaches the AI the same way it reaches you, and the turn-start check covers every major player. Independent powers and city-states are not penalised. In testing, an AI with no Iron kept building Warriors where, with Iron, it built Legions.

Under a locked value the AI is locked by the same data: an AI with no Iron can't train a Spearman, and its production queue never sticks on a locked unit (it builds something else). Independent powers and city-states ignore the gate by the game's own rule, so their spawns and armies are unchanged.

Compatibility

- Works with any mix of DLC. The unit tags only go on units that are in your game, so a unique unit from a DLC you don't own is skipped, and so is one a game update removes.
- Adds unit tags, modifiers and four game setup options. Below the locked values, every base-game row and file stays as the game has it; under a locked value, the mod sets the game's own `StrategicResource` column on the gated units and the game's six resource-minimum parameters, and makes the units a free unit upgrades into (every Modern unit with Modern Supply Chains on) grant-only behind their resource.
- Another mod that sets that column or those parameters (a locally installed gameplay overhaul does, with minimums of 2 to 6) keeps the engine's rule by its own numbers, and this mod's lock texts step back to their plain form rather than promise an unlock at one that does not hold. Turn one of the two off to get a single rule.
- The unit panel cell comes from a wrapper around the panel's own refresh, so mods that change the unit panel keep working. The production line goes in the slot the game keeps for mods on each production item, and the resource line into the text the game's description component draws; neither replaces a screen or a tooltip. Tested with Readable Tooltips, QD Improved Plot Tooltip, Building Location Advisor, City Hall, Better Commerce Screen UI and Mod Commons.
- The notification is an ordinary game notification of the mod's own type, and the callout an item in the game's own tutorial manager, so mods that restyle the notification train or the tutorials show them as they show the game's (tested with Mav's UI Overhaul, Nasuellia Popups and Nasuellia Diplomacy Screens).
- Mods that rewrite the game's per-resource bonus leave the penalty alone; the curve past six follows the base game's unit classes (read from code, not tested).
- Ships the Tower Settings Keeper, which runs first and keeps every mod's settings readable after a restart. The mod itself stores one flag: whether the first-time callout has shown.

More detail, by kind of mod: docs/compatibility.md.

Saves

The setup options and the unit data are fixed when a game is created. A game started with the mod needs it to load. If you add the mod while a game is under way, it takes effect from the next new game.

Install

Subscribe on the Steam Workshop, or copy the `strategic-resources-that-matter` folder into `~/Library/Application Support/Civilization VII/Mods/` (macOS) or `%LOCALAPPDATA%\Firaxis Games\Sid Meier's Civilization VII\Mods\` (Windows), and enable it under Add-ons.

Known limits

- Single-player only. The turn-start check is a local call, not a network request, so in a multiplayer game the mod switches it off and says so in the game's log. There, a lost resource brings its penalty back only once that player gains some resource.
- A loss shows one turn later, not at once.
- The first-time callout follows the game's options: with tutorials and advisor warnings both off it never shows. The game marks it seen in that save anyway, so turning warnings on later shows it in the next new game. In multiplayer neither the notification nor the callout shows.
- The curve stops at 20 of a resource.
- Under a locked value, a unit already in the production queue when the resource is lost keeps its progress and is completed: the game checks the resource when a unit is queued or bought, not when it finishes. Units it would have bought or queued after the loss are locked.
- The commander's Upgrade All is the game's own command, and its check doesn't read the resource: the mod greys the button and names the reason when every unit in the army is locked, and lists the locked ones when some can go. The engine refuses the locked units' upgrades either way; the AI's Upgrade All goes through the same engine check.
- The lock callout, like the unsupplied one, follows the game's tutorial options and shows once per profile.

How it works

- `data/`: each unit that needs a resource gets one tag, `SR_SUPPLY_<NAME>` (a second one with Modern Supply Chains on, from `sr-modern-chains.sql`). Per age and setup value, every major player's units get the combat penalty and their cities the production cut, each cancelled by an equal bonus while the player has one. The curve is a set of +1 bonuses behind higher counts. The files are written by `scripts/build-data.mjs` from the rules in `scripts/gate-map.mjs`.
- `ui/sr-recheck.js`: at the start of every turn, gains and drops one off-map unit of a resource nothing else reads, for every major player. Gaining a resource is what makes the game re-check a player's requirements at the next turn roll.
- `ui/sr-supply-lost.js`: the notification, raised from the same turn start for the local player when a resource held at the last turn start is gone. The notification type and its icon are in `data/sr-notification*.xml`.
- `ui/sr-callout.js`: the first-time callout, an item in the game's tutorial manager at the advisor-warning level.
- `ui/sr-unit-panel.js`: the status cell on the unit panel.
- `ui/sr-production-line.js`: the line on a production item, in the game's `ProductionModInfo` slot.
- `ui/sr-army-upgrade.js`: the commander's Upgrade All under a locked value, read against each packed unit's own upgrade check.
- `ui/sr-resource-line.js`: the line in a strategic resource's description, added where the game's `Stylize` component draws that text.
- `ui/sr-registry.js`, `ui/sr-player-supply.js`, `ui/sr-supply.js`: the shared registry wrapper, engine reads and pure rules the lines, the notification and the callout use, so they never disagree.
- `ui/settings-keeper.js`: the Tower Settings Keeper, unchanged.

The evidence for each of these, run by run, is in docs/verification-runs.md.

Development

Typed JavaScript checked by `tsc`, ESLint with size limits, and Node tests; no build step. See CONTRIBUTING.md.

```
npm install
npm run verify        # type check, lint, tests
npm run data          # rewrite data/ from the gate map
node scripts/deploy.mjs
```

License

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